

1 - Operational Pattern Definitions & Explanation of Exceptional Cases

Pattern	Logic (Downtime vs MTTR)	Included in Downtime/Breach Stats?
Data Error - Invalid Downtime	dt < 0	❌ Excluded — negative value is invalid
Outlier Downtime - Needs Data Audit	dt > IQR threshold	❌ Excluded — suspected unit/entry error
Data Anomaly - Impact Without Downtime	dt = 0, MTTR = 0, txns > 0	✅ Yes
Self-Resolved (Auto Recovery)	dt > 0, MTTR = 0	✅ Yes
Successful Resolution	dt > 0, MTTR ≥ dt	✅ Yes
Process Delay - Extended MTTR	admin_delay > IQR threshold	✅ Yes (Downtime value still valid)
Logging Error - Premature Closure	dt > 0, MTTR < dt	✅ Yes (Downtime value still valid; only MTTR unreliable — excluded from Admin Delay analysis only).
False Alert - No Actual Outage	dt = 0, MTTR > 0	✅ Yes (no impact on Downtime SUM)

Data Error: Excluded from all aggregate statistics because the Downtime value is not trustworthy (negative). These are data logging errors, not genuine incidents.

Outlier Downtime: Excluded from aggregate statistics and reviewed separately in the Outlier Deep-Dive. These values may skew results, but could reflect genuine, severe incidents — not necessarily data errors.

Logging Error - Premature Closure: MTTR < Downtime indicates the ticket was closed before the service was actually restored — a suspected discrepancy in ticket-closure timing. Excluded from aggregate statistics to keep reported figures consistent.

Outlier thresholds: All anomaly thresholds are calculated automatically using the IQR method (Interquartile Range, 1.5x multiplier) rather than a fixed cutoff, ensuring the threshold adapts to the actual distribution of the data.

2 - Data Excluded from Aggregate Stats

Is Valid For Stats	Operational Pattern	# Incidents	% of Incidents
True	Successful Resolution	931	62.07%
True	Self-Resolved (Auto Recovery)	182	12.13%
False	Logging Error - Premature Closure	96	6.40%
True	Process Delay - Extended MTTR	95	6.33%
False	Outlier Downtime - Needs Data Audit	77	5.13%
True	Data Anomaly - Impact Without Downtime	47	3.13%
True	False Alert - No Actual Outage	43	2.87%
False	Data Error - Invalid Downtime	29	1.93%
Total		1500	100.00%

Data Exclusion Rule

- Data Error - Invalid Downtime are excluded from all pages
- Outlier Downtime - Needs Data Audit are excluded from all pages except the Outlier Deep-Dive page, where they are the subject of analysis.
- Logging Error - Premature Closure accounts for 6.4% of records — since its Downtime value remains valid, it is kept in all other pages, and excluded only from the Admin Delay analysis, where MTTR reliability matters.
- Self-Resolved (Auto Recovery) records (12.13%) are excluded from Admin Delay only, since MTTR = 0 means no human action occurred and the metric is not meaningful for these cases.

Recommendation

- Data Error - Invalid Downtime needs to re-check with database & logging team
- The rest excluded data is addressed in Outlier Deep-Dive page

Choose Date

1/1/2024



12/31/2025



Operational_Pattern Filter

- ☒ Data Anomaly - Imp...
- ☐ Data Error - Invalid ...
- ☒ False Alert - No Act...
- ☒ Logging Error - Pre...
- ☐ Outlier Downtime - ...
- ☒ Process Delay - Ext...
- ☒ Self-Resolved (Auto ...
- ☒ Successful Resolution

Incidents

1.4K

Excluded: Data Error, Outlier Downtime

Total Downtime (mins)

29.94K

Total Financial Loss

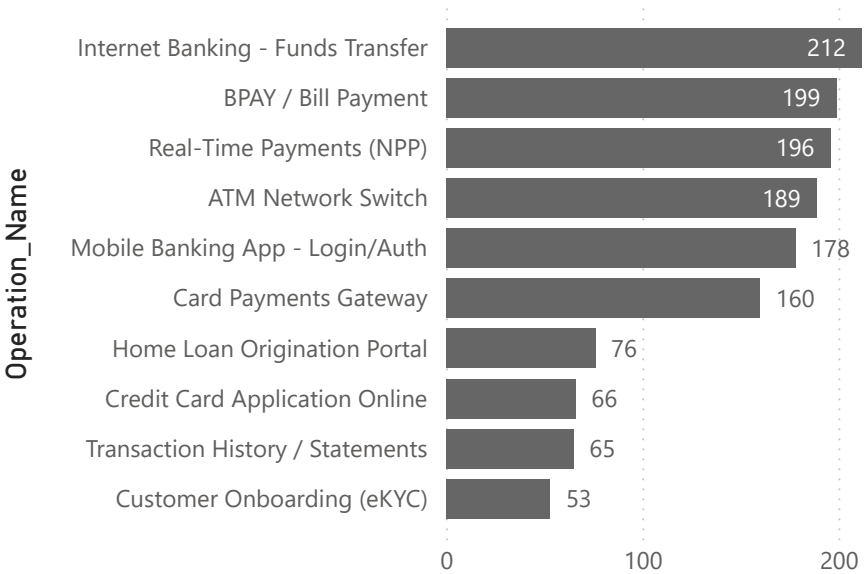
50.8M

2% of incidents missing finance info

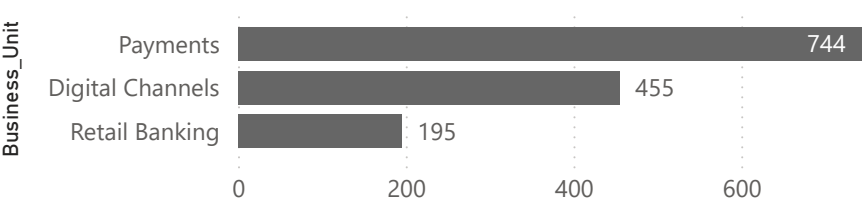
% Breached Operations Tier1

100.0%

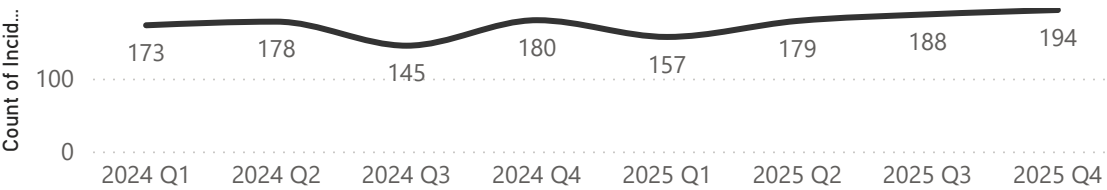
Incidents By Operation



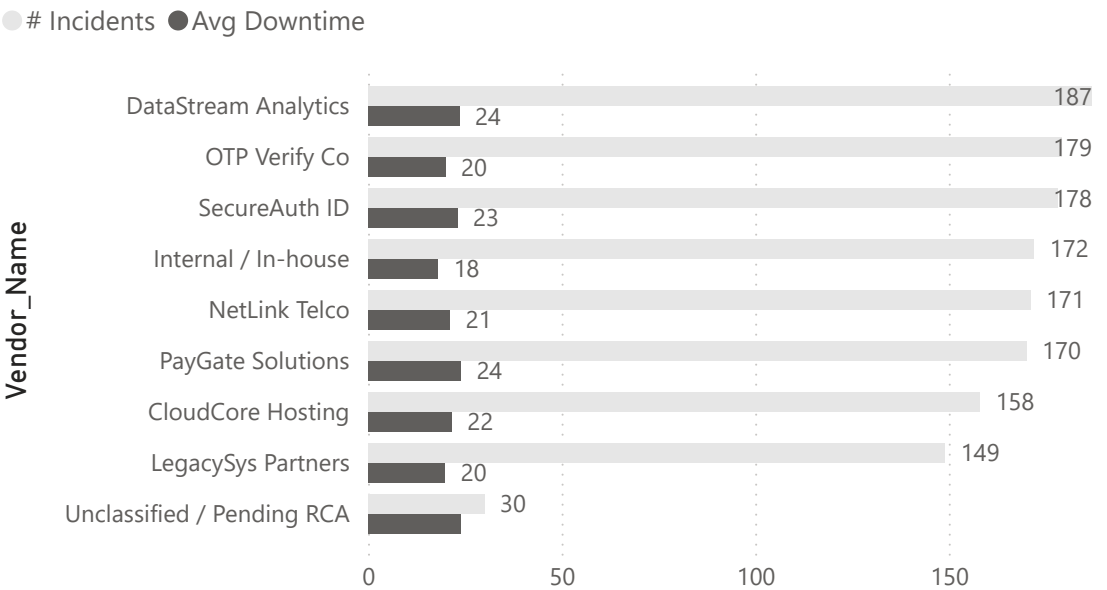
Incidents By BU



Quarterly Incidents



Incidents By Vendor



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Criticality_Level

☒ Tier 1

☐ Tier 2

Operational_Pattern
Filter

☒ Data Anomaly - Imp...

☐ Data Error - Invalid ...

☒ False Alert - No Act...

☒ Logging Error - Pre...

☐ Outlier Downtime - ...

☒ Process Delay - Exte...

☒ Self-Resolved (Auto ...

☒ Successful Resolution

PAGE 3: Impact Tolerance Breach Monitor

Breached Ops

6

Excluded: Data Error, Outlier Downtime, Tier2

% Breached Ops

100.0%

Incident Breach Rate (%)

65.7%

Incident Breaches

745

Executive Findings:

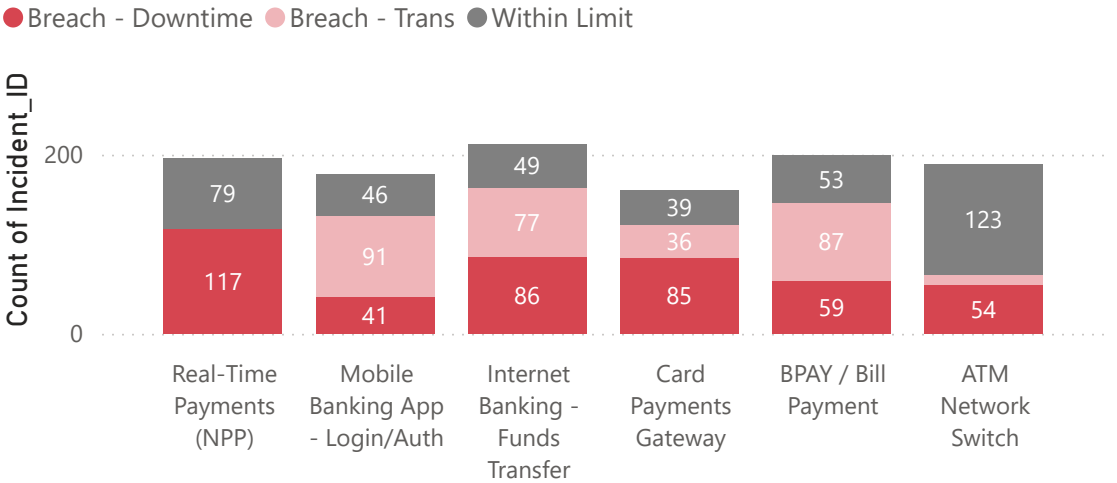
- **High Breach Rate:** 100% of Tier 1 operations and 65.7% of total incidents breached tolerance limits.
- **Primary Breach Drivers:** Downtime is the main driver for *Real-time Payments*, *Internet Banking*, and *Card Payments Gateway*, whereas *Mobile Banking* and *BPAY* mainly breach failed transaction limits.
- **Severity Insight (Total Downtime):** While *Real-time Payments* has the highest breach downtime incident volume (117), **Card Payments Gateway**, **BPAY**, and **Mobile Banking App** contribute the vast majority of total breach downtime (~27K, ~17K and ~18K minutes), posing the highest operational risk.
- **Best Performer:** *ATM Network Switch* recorded the lowest breach rate (~35%).

Recommendation:

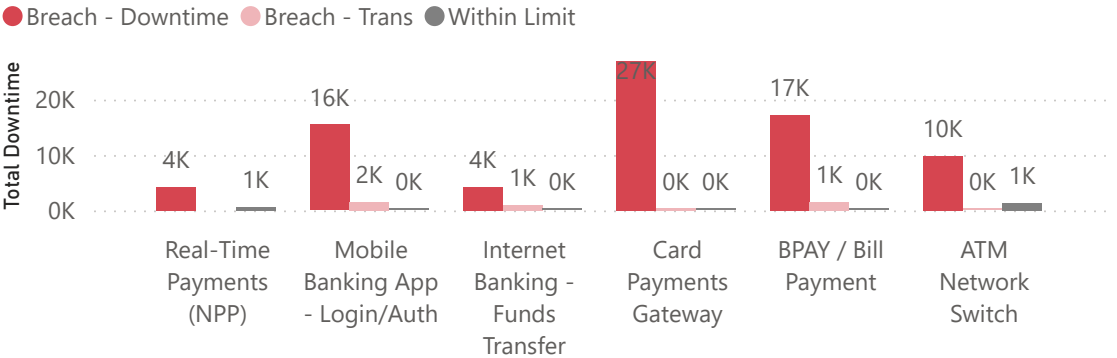
- Prioritize operational resilience upgrades for **Card Payments Gateway** and **BPAY** to control severe downtime, while optimizing transaction handling for **Mobile Banking App**.

Incident_ID	Operation Name	DownTime
INC-000001	Internet Banking - Funds Transfer	47.00
INC-000002	Real-Time Payments (NPP)	40.00
INC-000004	Card Payments Gateway	37.00
INC-000005	Internet Banking - Funds Transfer	22.00
INC-000006	Mobile Banking App - Login/Auth	5.00
INC-000008	ATM Network Switch	19.00
INC-000009	Mobile Banking App - Login/Auth	16.00
INC-000010	Internet Banking - Funds Transfer	0.00
INC-000011	Internet Banking - Funds Transfer	9.00
INC-000012	BPAY / Bill Payment	18.00
Total		23,965.00

Incidents By Operation & Breach



Total Downtime Of Breach By Operation



Choose Date

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Operational_Pattern Filter

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- ☐ Self-Resolved (Auto ...
- ☒ Successful Resolution

PAGE 4: Admin Delay

Incidents

1.1K

Admin Delay Time (mins)

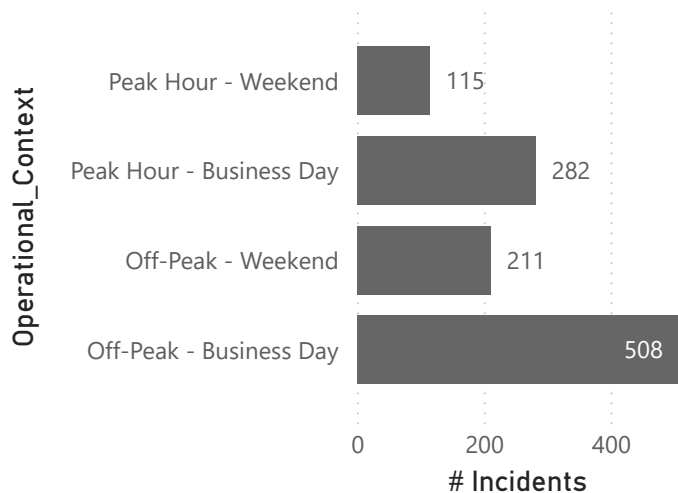
50.58K

P95 Admin Delay

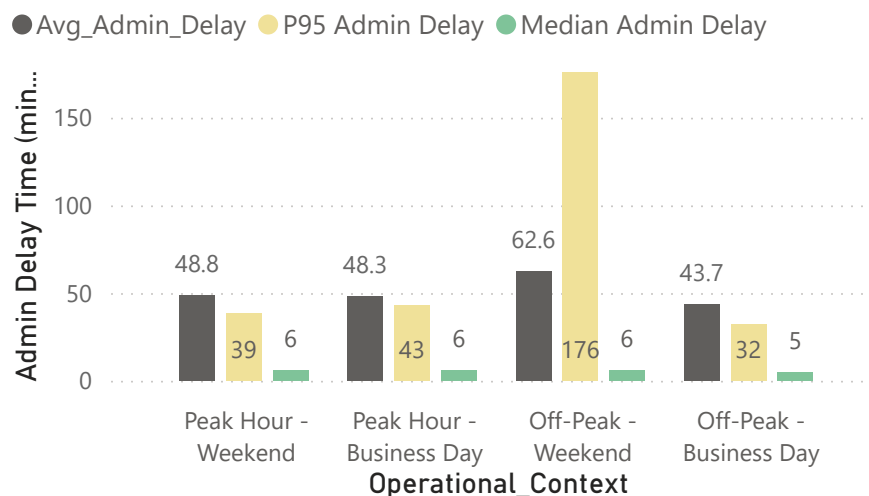
41.0

Excluded: Data Error, Logging Error, Outlier Downtime and Auto Healing

Incidents By Operational Context



Admin Delay By Operational Context



Core Performance vs. Outlier Impact

- Routine Efficiency:** Median delay remains consistently low at 5–6 mins across all operational contexts, showing prompt routine handling.
- Outlier Skewness:** High avg. delays (32–62.6 mins) indicate a right-skewed distribution driven by long-tail outliers rather than systemic operational failure.

Off-Peak Weekend Bottleneck

- Weekend Off-Peak Anomaly:** an extreme P95 delay of **176 mins** — over 4x higher than any other context
- Root Cause:** Weekend ticket handover, escalation workflows, or on-call coverage.

Recommended Actions

- Core workflow is sound, no change needed for routine cases (Median 5–6 mins)
- Auto-alert for tickets exceeding 40 mins, calibrated to business-day P95
- Review weekend on-call staffing for off-peak hours

Operational Pattern	Operational Context	# Incidents	Avg Admin Delay	P95 Admin Delay	Medium Admin Delay	Total Admin Delay
Process Delay - Extended MTTR	Off-Peak - Business Day	38	455.0	2,233.8	56	17,289
Process Delay - Extended MTTR	Off-Peak - Weekend	20	537.2	1,792.2	334	10,743
Process Delay - Extended MTTR	Peak Hour - Business Day	27	428.4	1,774.7	47	11,566
Process Delay - Extended MTTR	Peak Hour - Weekend	10	432.1	1,374.9	48	4,321
Successful Resolution	Off-Peak - Business Day	425	7.0	21.0	5	2,965
Successful Resolution	Peak Hour - Business Day	245	6.4	19.6	5	1,574
Successful Resolution	Peak Hour - Weekend	91	6.7	19.0	5	610
Successful Resolution	Off-Peak - Weekend	170	6.8	17.5	6	1,158
Total		1116	49.0	41.0	6	50,583

Choose Date

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- Operational_Pattern Filter
- ☒

 Data Anomaly - Imp...
- ☐

 Data Error - Invalid ...
- ☒

 False Alert - No Act...
- ☒

 Logging Error - Pre...
- ☐

 Outlier Downtime - ...
- ☒

 Process Delay - Exte...
- ☒

 Self-Resolved (Auto ...
- ☒

 Successful Resolution

Incidents

1.4K

Total Financial Loss

50.8M

Avg Financial Loss

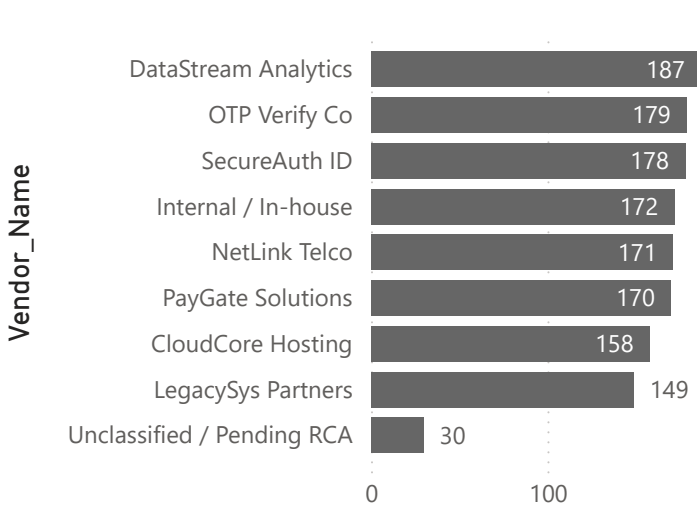
38.7K

P95 Downtime

56.0

Excluded: Data Error, Logging Error, Outlier Downtime

Incidents By Vendor



Vendor Risk Score:

- **Formula:** Min-Max Normalized Weighted Score
- **Risk Score** = (Financial_Loss_Score × 40%) + (Incidents_Score × 20%) + (Avg_Downtime_Score × 25%) + (P95_Downtime_Score × 15%)
- ***Where Score** = (Current - Min) / (Max - Min)
- **Final Output:** High Risk → 100 | Low Risk → 0

Executive Findings:

- **Top Financial Risk:** **DataStream Analytics** causes the highest loss (**8.45M**, avg 48K/incident) and most incidents (187).
- **Top Operational & Downtime Risk:** **PayGate Solutions** ranks worst overall with high losses (7.04M) and the highest downtime (Median 21m, P95 61.5m). SecureAuth ID rank second (risk score 35), downtime severity (P95 60.3 mins) pushed it above Data Stream Analytics
- **Gold Tier Failure:** Top 3 loss contributors (*DataStream*, *PayGate*, *SecureAuth*) all hold **Gold SLAs**, failing to justify their tier.
- **Cost-Effective Option:** **OTP Verify Co** shows strong efficiency (Median 15m, low Avg Loss 32.7K) despite being Under Review. Bronze and Silver Tier have noticeably better (lower) risk score than Gold and Platinum Tier.

Recommendations

- **Renew & Upgrade:** Formalize contract with **OTP Verify Co**.
- **Audit & Enforce Penalties:** Review SLAs and apply penalty clauses for underperforming Gold vendors (**DataStream**, **PayGate**, **SecureAuth**).

Vendor Name	Contract Status	SLA_Tier	# Incidents	Vendor Risk Score	Medium Downtime	Financial Loss	Avg Financial Loss	Avg Downtime	P95 Downtime
DataStream Analytics	Active	Gold	187	32	16.0	8,447,239	47,995.7	23.6	56.0
PayGate Solutions	Active	Gold	170	40	21.0	7,040,145	42,927.7	23.9	61.5
SecureAuth ID	Active	Gold	178	35	20.0	6,679,929	39,761.5	23.0	60.3
CloudCore Hosting	Active	Platinum	158	24	17.5	5,888,174	38,484.8	21.6	56.5
LegacySys Partners	Expired	Bronze	149	8	17.0	5,402,781	38,317.6	19.9	49.2
NetLink Telco	Active	Silver	171	16	18.0	6,005,637	37,535.2	21.1	52.0
OTP Verify Co	Under Review	Silver	179	14	15.0	5,690,781	32,705.6	20.0	54.0
Internal / In-house	Active	N/A	172	2	14.0	5,153,657	32,010.3	18.0	50.9

30 incidents (2% of total) lack classified vendor information, and require root cause analysis to identify the responsible vendor

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Operational_Context

- ☐ Off-Peak - Busin...
- ☐ Off-Peak - Week...
- ☐ Peak Hour - Busi...
- ☐ Peak Hour - Wee...

Operation_Name

- ☐ ATM Network Sw...
- ☐ BPAY / Bill Payme...
- ☐ Card Payments G...
- ☐ Credit Card Appli...

Vendor_Name

- ☐ CloudCore Hosti...
- ☐ DataStream Anal...
- ☐ Internal / In-house
- ☐ LegacySys Partners

PAGE 6: Outlier Deep-Dive

Data Anomaly Incidents

47

Impacted Trans
11.87K

Outlier Downtime Incidents

77

Financial Loss
144M

Downtime (mins)
64.99K

Est. Financial Loss (Unverified)

Logging Error Incidents

96

% Premature Closure Rate
6.4%

Executive Findings:

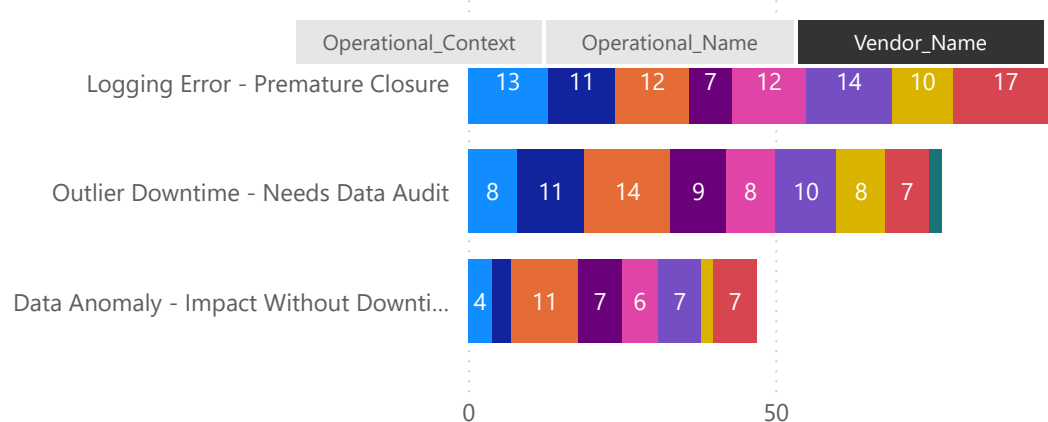
- **Financial Loss Anomaly:** Abnormal incidents represent just **15% (220 cases)** but account for **\$144M** in loss (vs. **\$50.8M** for resolved cases). Further verification is needed to pinpoint the root cause, whether it indicates data distortion or genuine financial loss.
- **Off-Peak Vulnerability:** 65–74% of outlier incidents occur during off-peak hours, signaling weak off-peak monitoring or slow escalation response.
- **Vendor Non-Compliance:** SecureAuth ID ranks 1st in Logging Errors despite ranking only 3rd in total incidents—pointing to potential premature ticket closures to artificially maintain MTTR SLAs.
- **High-Impact Operations:** BPAY and ATM Network Switch drive the majority of Outlier Downtime and represent 45% of **144M financial loss**.
- **Internal Blind Spots:** Internal systems lead Data Anomalies (11 cases), revealing unmonitored silent outages in core applications.
- **Impact Scaling:** Longer downtime directly correlates with higher financial loss.

Recommended Actions:

- **Process & Vendor Quality:** Mandate logging quality improvements for SecureAuth ID, OTP Verify Co, and NetLink Telco.
- **Root Cause & Alerting:** Review repair workflows and RCA for BPAY and ATM operations to upgrade on-call alerting.
- **Automated Data Rules:** Strengthen automated data validation rules and monitoring to instantly detect internal silent outages.

Outlier Incidents By Operations / Vendors / Operational Context

- CloudCore Hosting
- DataStream Analytics
- Internal / In-house
- LegacySys Partners
- NetLink Telco
- OTP Verify Co
- PayGate Solutions
- SecureAuth ID
- Unclassified / Pending RCA



Outlier Downtime Severity: Downtime vs. MTTR

